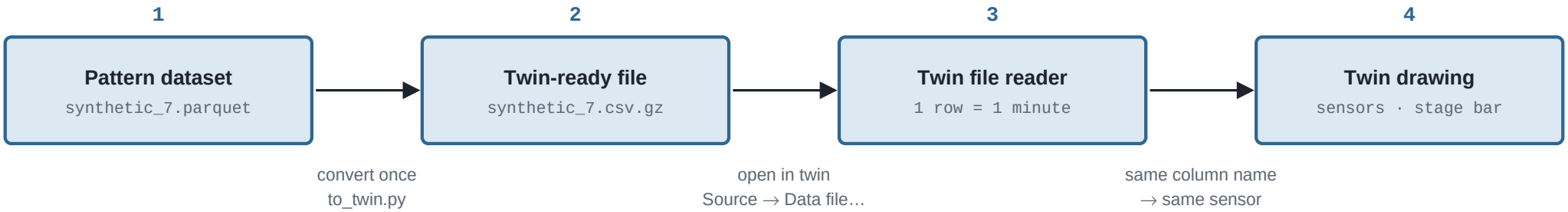
Playing a made-up plant day in the twin

You convert the dataset into a CSV file once. Then in the twin you choose **Source → Data file…** and open it. Each minute of the file then takes the place of the twin's own simulated minute. Every column lands on the sensor that has the same name.

This means you can watch any synthetic day on the plant drawing. Every sensor reading, plot, trend and the stage bar moves as if the plant were running. Nothing in the twin had to change: it already reads data files. The pattern datasets only needed one small conversion step.

Think of the twin as a player piano. Normally it plays its own tune (the simulation). Feeding it a data file swaps in a recorded roll, and every key it presses is the sensor named on that line of the roll.



The data goes one way only: file to twin. The conversion (step 2) is the only new piece. Steps 3 and 4 are the twin's existing "Data file" playback.

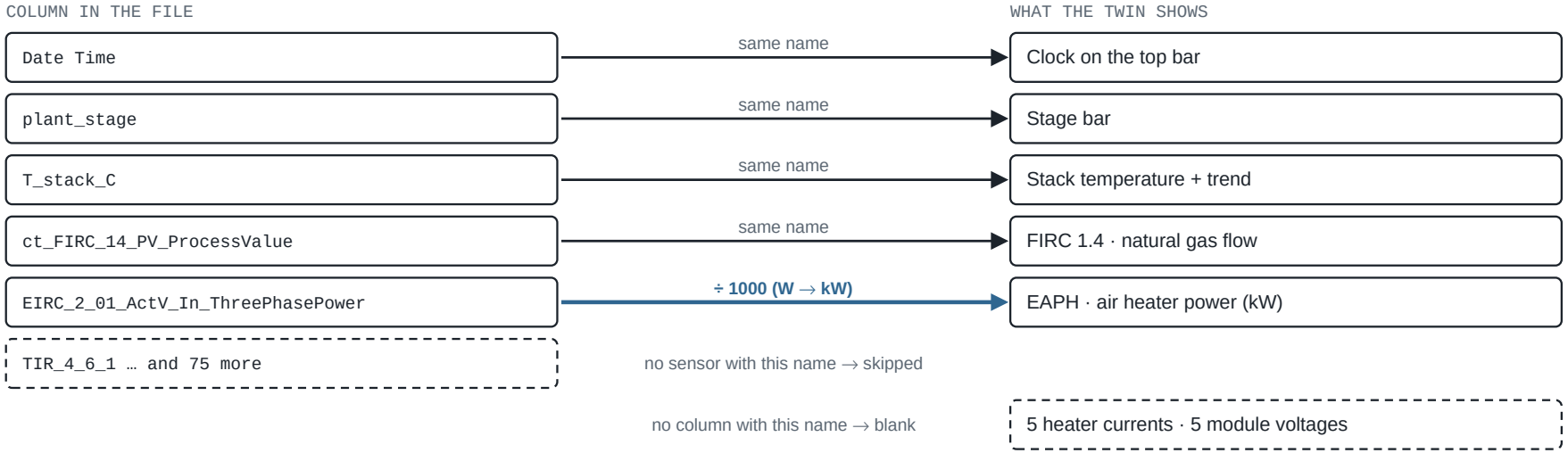
Question	Answer
Do the sensor names match?	Yes. 101 of the twin's 111 sensors have a column with exactly the same name (page 2).
Do the stage names match?	Yes. The file's plant_stage column uses the twin's stage names, so the stage bar follows the file.
Why convert at all?	The twin runs in a web browser, which cannot open .parquet files. The conversion also changes heater power from watts to kilowatts.
Was it tested?	Yes. All 3 datasets loaded through the twin's own reader (page 3).

If you remember one thing: convert once, then pick Source → Data file…. The twin puts each column on the sensor with the same name.

How each column finds its sensor

The twin reads the first line of the file, which holds the column names. For each name it looks for a sensor with exactly the same name. If it finds one, that column's numbers go to that sensor. If not, the column is skipped.

No mapping table is needed. The pattern filler copies real plant minutes, so its columns keep the plant's own tag names. The twin was built on those same names.



Five sample rows out of 101 matches. The blue line is the only column whose numbers change on the way. Dashed boxes have no partner and do nothing.

Group	How many	What happens
Twin sensors with a matching column	101 of 111	Show the file's values
Twin sensors with no column	10	Stay blank: five heater currents (IIR 3.1.1, 5.1.1–5.1.3, 6.1.1) and five module voltages (TotalVoltage M1–M5)
Readings the twin works out itself	9	Total power, current and voltage, and six efficiencies, from the five strings' current and voltage and the gas flow
File columns with no twin sensor	76	Skipped (for example the 36 individual stack thermocouples TIR 4.6.x)

The three things the conversion changes

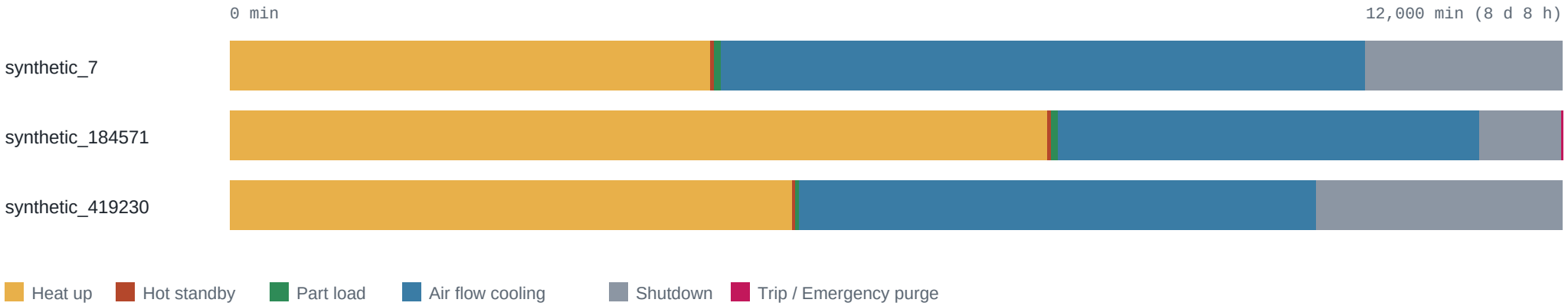
Item	In the parquet file	In the twin-ready file	Why
Air heater power	21,146 W	21.15 kW	The twin's EAPH box is in kilowatts
Missing reading	–1	(blank)	The plant logger writes –1 for "no value"
Stage "No data"	No data	(blank) → twin's own stage rules	A few filler minutes had no stage

If you remember one thing: the link between file and twin is just the column name. 101 of 111 sensors match as they are. Only heater power needs a unit change.

Step by step, and what we checked

Step	What to do
1 Make or pick a dataset	Pattern app or generator + filler → 04_synthetic_data/datasets/<run folder>/synthetic_<seed>.parquet
2 Convert it once	python "04_synthetic_data/to_twin.py" (all datasets; or add one file path). Writes a .csv.gz next to each parquet.
3 Open the twin	Double-click 05_twin_2d/app.bat (or start.bat).
4 Source → Data file…	Top right, beside the clock. Pick the .csv.gz. The clock shows "Loading … min" while it reads.
5 Watch and move around	Play / Pause and speed run through the file; the slider jumps to any minute; click a part to see its sensors; Save data exports.
6 Source → Twin	The simulation comes back exactly where you left it.

Locked during playback: the knobs, stage buttons and Start plant. The file decides what the plant does, so the twin's physics does not run. You are watching a recording, not a simulation.



Minutes per stage as the twin names them. Its "Heat up" includes the file's Hot standby hold, and its "Hot standby" is the file's Fuel switch.

Dataset	Minutes read	Part load (min)	Power in part load	Readings filled	Result
synthetic_7	12,000	63	e.g. 3.2 kW	110 of 120	Pass
synthetic_184571	12,000	61	e.g. 4.6 kW	110 of 120	Pass
synthetic_419230	12,000	40	e.g. 3.2 kW	110 of 120	Pass

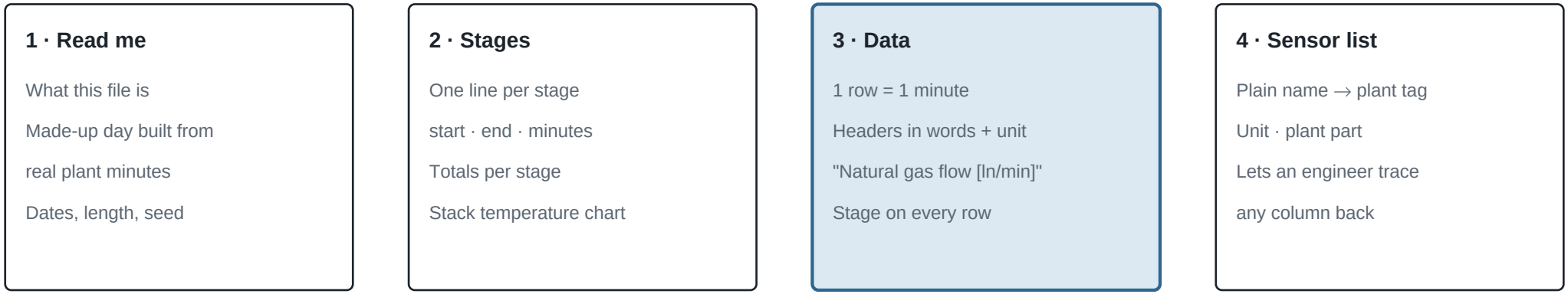
Limits: 10 readings stay blank (heater currents, module voltages). The heater lamp is decided from air heater power alone (above 0.2 kW). The test ran the twin's reader code in Node, not by clicking through Edge.

If you remember one thing: run to_twin.py once. Then Source → Data file… plays the dataset, and Source → Twin brings the simulation back.

Giving a dataset to someone non-technical

Hand it over as **one Excel workbook per dataset**, with four sheets: a read-me, a stage summary, the minute data with plain column names and units, and a sensor list.

Everyone can open Excel by double-clicking. The files we use internally (parquet, zipped CSV) need Python or extra steps. Their column names are plant tags like ct_FIRC_14_PV_ProcessValue, which mean nothing to a newcomer.



The four sheets of the workbook. The highlighted Data sheet is the dataset itself. The other three explain it.

Format	Opens with	Fit for a non-technical reader
.parquet (what the filler saves)	Python only	No
.csv.gz (what the twin reads)	Unzip first, then Excel; tag names, no units	No
Plain .csv	Excel; tag names, no units, no explanation	Partly
Excel workbook, 4 sheets	Double-click; plain names, units, read-me	**Yes**
Twin playback	Needs the twin app on their PC	Show it to them; don't send it

Size is fine for Excel: 12,000 minutes × about 180 columns. Plain names exist today for the twin's 111 sensors; the other columns keep their tag names or can be left out.

If you remember one thing: send an Excel workbook with plain names, units and a read-me sheet. Keep the parquet and csv.gz files for the tools.